

Bayesian Literature Review

ar007 · 19 May 2026

Paper	Year
Ma, Beck, Latham & Pouget — <i>Bayesian Inference with Probabilistic Population Codes</i>	2006
Pouget, Dayan & Zemel — <i>Inference and Computation with Population Codes</i>	2003
Fiser, Berkes, Orbán & Lengyel — <i>Statistically Optimal Perception and Learning: From Behavior to Neural Representations</i>	2010
Orbán, Berkes, Fiser & Lengyel — <i>Neural Variability and Sampling-Based Probabilistic Representations in the Visual Cortex</i>	2016
Aitchison & Lengyel — <i>The Hamiltonian Brain: Efficient Probabilistic Inference with Excitation-Inhibition Networks</i>	2016
Echeveste, Aitchison, Hennequin & Lengyel — <i>Cortical-like Dynamics in Recurrent Circuits Optimized for Sampling-Based Probabilistic Inference</i>	2020
Padamsey, Katsanevaki, Dupuy & Rochefort — <i>Neocortex Saves Energy by Reducing Coding Precision during Food Scarcity</i>	2022

Terms to know first

The papers above share a small vocabulary that runs across all of them. If a term feels hazy, look it up before reading — they are not redefined in each paper.

Bayesian fundamentals.

- *Prior, likelihood, posterior.* $p(\theta)$, $p(x | \theta)$, $p(\theta | x) \propto p(x | \theta)p(\theta)$. The posterior is what every paper here claims the brain represents in some form.
- *Generative model.* The forward model the brain is assumed to have inverted: latent causes $\rightarrow$ sensory observations.
- *Recognition / inference.* The inverse problem — recovering $p(\theta | x)$ from x — performed approximately by the circuit.
- *Marginalisation.* Integrating out nuisance variables. Cortical circuits are claimed to do this in different ways depending on whether the representation is parametric or sample-based.

Two flavours of neural representation of probability. This is the central split in the reading list and the two camps disagree on it.

- *Parametric (probabilistic population code, PPC).* The instantaneous population firing rate $\mathbf{r}$ encodes the parameters of a distribution over θ — for an exponential-family code, $p(\theta | x) \propto \exp(\mathbf{h}(\theta)^\top \mathbf{r})$. Posterior uncertainty is read off the population gain. Ma et al. 2006 and Pouget et al. 2003 are the canonical PPC papers.

- *Sampling-based*. The instantaneous activity $\mathbf{r}(t)$ is a sample from the posterior, $\mathbf{r}(t) \sim p(\theta \mid x)$. Uncertainty is read off the trial-to-trial variability over time. Fiser et al. 2010, Orbán et al. 2016, and the Lengyel-group papers (Aitchison & Lengyel 2016, Echeveste et al. 2020) are the canonical sampling-camp papers.
- *Variational inference (background)*. A third flavour — fit a parametric $q(\theta)$ to the posterior by minimising KL divergence. Worth knowing as the ML reference point, though none of these papers commit to it as the brain’s algorithm.

Population-code mechanics.

- *Tuning curve* $f_i(\theta)$ — neuron i ’s mean rate as a function of stimulus θ .
- *Fisher information* $I_F(\theta)$ — the precision a code carries about θ ; for an unbiased estimator, $\text{Var}(\hat{\theta}) \geq 1/I_F(\theta)$ (Cramér–Rao). Often discussed per spike, to compare codes at matched energy.
- *Linear Fisher information*. The piece of I_F accessible by a linear readout — what a downstream neuron with synaptic weights can actually extract.
- *Noise correlations* c_{ij} — trial-to-trial covariance between neurons at fixed stimulus. Information-limiting correlations are the slice of c_{ij} aligned with the tuning gradient $\partial f/\partial \theta$; they cap I_F no matter how many neurons you average.
- *Poisson variability*. Spike-count variance $\approx$ mean. The baseline noise model PPCs are built on.

Sampling-camp specifics.

- *Langevin dynamics*. Stochastic gradient on $-\log p(\theta \mid x)$ plus white noise — the simplest dynamical system whose stationary distribution is the posterior. The recurrent-circuit papers are versions of this idea with biological constraints.
- *Hamiltonian Monte Carlo (HMC)*. Sampling with momentum — propose moves along trajectories of a fictitious Hamiltonian, accept by Metropolis. Aitchison & Lengyel argue E/I circuits implement an HMC-like sampler with the inhibitory population playing the momentum role.
- *Inhibition-stabilised network (ISN)*. A recurrent E/I circuit whose excitatory subnetwork is *unstable* on its own and stabilised only by inhibition. Empirically common in cortex; the sampler papers exploit ISN dynamics for fast, well-mixed sampling.
- *Mixing time, autocorrelation time*. How quickly a sampler decorrelates from its current state — sets how fast a circuit can revise its uncertainty estimate. Echeveste et al. argue cortical-like transients arise from optimising for fast mixing.
- *Stochastic stability vs. transients*. A sampler must be globally stable but locally fluctuating. The non-trivial transient responses cortical circuits show are read as evidence of a sampler near, not at, equilibrium.

Coding-cost and metabolism.

- *Bits per spike*. Mutual information per spike between stimulus and response — the efficiency the energetics papers ask about.
- *Coding precision*. The width of the represented posterior; can be widened to save metabolic cost without changing the represented mean. Padamsey et al. 2022 show this

happens *in vivo* under food scarcity — directly tying the Bayesian and metabolic stories together.

- *Metabolic cost of a spike.* Sodium-pump ATP per action potential; the currency the precision–cost trade-off is denominated in.

Useful background that none of these papers stop to define.

- *Exponential family, sufficient statistics, natural parameters.* PPCs lean on this — the firing rates are the natural parameters of an exponential-family posterior.
- *KL divergence.* The asymmetric distance between distributions, used to score how good an approximate posterior is.
- *Stationary distribution of a Markov chain.* The distribution the chain converges to and samples from at long times. The sampling-camp claim is that the cortical chain’s stationary distribution is the posterior.